MIL-T-3462D
7 June 1976
SUPERSEDING
MIL-T-003462C (DSA-DM)
26 November 1973 and
MIL-T-3462B
14 December 1971

MILITARY SPECIFICATION

THERMOMETER, SELF-INDICATING, BIMETALLIC, 5-1/2 Inch

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers a dial type thermometer for use by personnel of the Veterinary Service in inspection of food products procured for military consumption and in determining the temperature of shell eggs.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-P-116	Preservation-packaging, Methods of.

FSC 6685

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STANDARDS

Federal

Fed. Std. No. 66	Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Thermometer. The dial frame, stem, and adjustment nut of the thermometer shall be fabricated from corrosion-resisting steel conforming to Fed. Std. No. 66, AISI 300 series.

3.1.2 Dial crystal. The dial crystal shall be fabricated from glass or plastic and shall be optically clear and free of cracks. The dial crystal shall show no evidence of discoloration or deformation when tested as specified in 4.4.4.

3.1.3 Carrying case. The carrying case shall be fabricated from hard rubber or rigid plastic. Commercial colors are acceptable for the finish of the case.

3.1.4 Pocket clip. The pocket clip shall be fabricated from either corrosion-resisting steel or metal suitably treated to resist corrosion.

3.2 Components.

3.2.1 Thermometer. Shall be a dial type, self-indicating, bimetallic thermometer with a circular face. The thermometer shall conform to Figure 1 for design and dimensions, and shall consist of a metal stem attached to a dial frame. The stem shall be a housing for a bimetallic heat-sensitive element. As indicated in Figure 1, the thermometer shall be fitted with a calibration nut (see 3.3). The

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stem shall be straight and shall be positioned perpendicular to the center of the base of the dial frame. The dial frame shall be a housing for a dial face and a black dial pointer with a protective crystal covering (see 3.1.2) that is securely sealed to the dial frame to render the thermometer watertight. The dial face shall have a white or silver background with black graduations, black numerals. Graduations shall extend around the circumference of the dial face as indicated in Figure 1. The graduations shall have a range from 0 to 220° Fahrenheit (°F.) and shall be in increments of 2°. Each 10° mark shall be longer than intervening marks; at least each 20° mark shall be indicated with Arabic numerals. The number and graduation marks shall be clear and shall not be ragged or irregular.

* 3.2.2 Carrying case. The carrying case with pocket clip shall be constructed in accordance with Figure 1 for design and dimensions. The thermometer shall be capable of being inserted into the clip end of the carrying case, and then held in place by the friction of either a clip support within the cavity of the case, or the cavity of the case when an adjustable spring clip is supplied. The carrying case shall also function as a handle for the thermometer by means of the thermometer stem being inserted either through an exposed loop of the pocket clip or under an adjustable spring clip. The thermometer shall be held in place when inserted in the handle of the carrying case. The clip shall be securely attached to the case. The case shall be rigid and shall be free of warping, cracks, breaks, sharpness and burrs. The color of the case shall be permanent.

* 3.3 Performance. The thermometer shall indicate temperatures by means of a rotary action of the bimetallic element turning a shaft that is connected to the dial pointer. When necessary, scale adjustments shall be capable of being made without disassembling the thermometer, with adjustments being made by gripping the nut with a wrench and then turning the dial frame to the proper position.

* 3.3.1 Accuracy. All thermometers shall be properly adjusted to indicate the correct temperature. The thermometer readings shall be accurate within 2°F. (one increment) when tested as specified in 4.4.1.

* 3.3.2 Speed of response. The thermometer shall respond rapidly to temperature changes. When tested as specified in 4.4.2, the time for the thermometer reading to rise from 50° to 180°F. shall not exceed 10 seconds.

3.3.3 Stability. The thermometer reading shall not change more than 1°F. when tested as specified in 4.4.3.

3.4 Identification marking. Each thermometer shall be legibly marked with the model number and name or registered trademark of the manufacturer. As an alternate, the model number may be included in the instructions for use (see 3.5).

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3.5 Instructions for use. Each thermometer shall be provided with printed instructions for use and care of the thermometer.

3.6 Caution statement. A caution statement shall be included with each thermometer to indicate that the thermometer will be accurate within 2°F. unless damaged by shock or temperature extremes, but can be adjusted to a 1°F. accuracy at any desired point by carefully calibrating at that specific temperature. The caution statement may be part of the instructions of 3.5 or an insert.

3.7 Workmanship. Workmanship shall be first class throughout. The thermometer and carrying case shall be free from defects which detract from their appearance or may impair their serviceability. The thermometer shall be supplied without sharp edges, protrusions, burrs, pits, rough areas or distortion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the steel or dial crystal, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105. The unit of product for inspection purposes shall be the thermometer and carrying case.

Table I For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table II and MIL-STD-105. The unit of product shall be the thermometer.

Table II For tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Accuracy	3.3.1	4.4.1	S-4	**
Speed of response	3.3.2	4.4.2	S-2	**
Stability	3.3.3	4.4.3	S-2	**
Dial crystal	3.1.2	4.4.4	S-2	**

* Acceptance number shall be zero.

4.3 Examination. The thermometer shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

Table III Classification of defects

Category	Defect
Major	
101	Component part missing.
102	Thermometer not free of sharp edges.
103	Thermometer not free of protrusions.
104	Dial face or crystal not clear.
105	Dial face or crystal not free of cracks.
106	Dial face not secured to frame.
107	Graduation markings not complete, correct and legible.

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Classification of defects (cont'd.)

Category	Defect
Major	
108	Stem not secured to head.
109	Stem not straight.
110	Stem not perpendicular to base of dial frame.
111	Calibration nut not properly secured to thermometer.
112	Clip cannot be secured to thermometer.
113	Case not free of warping or not sufficiently rigid.
114	Case not free of cracks and breaks.
115	Crystal not adequately secured to dial frame.
116	Thermometer not capable of being readily inserted into or removed from case.
117	Thermometer not free of moisture on dial face.
Minor	
201	Metal surfaces not free of pits and burrs.
202	Thermometer not free of rough areas.
203	Thermometer not free of scratches.
204	Identification marking not complete, correct, legible, and permanent.
205	Dial face not free of scratches and marks.
206	Dial face not free of warping.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The thermometer shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4. Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The thermometer shall be tested against a standard thermometer at 32°F., 122°F. and 212°F. Calibration of the standard thermometer shall be traceable to the National Bureau of Standards.

4.4.2 Instant recording. The stem of the thermometer shall be immersed in a liquid bath which has a temperature of 32°F. until the thermometer registers 32°F. Thermometer shall be removed from this 32°F. bath, and the stem of the thermometer shall be immediately immersed in a 212°F. bath. The time elapsing between readings of 50° and 180°F. shall be determined.

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4.4.3 Stability. A temperature reading shall be obtained by immersing the stem of the thermometer in a liquid of any temperature between 2° and 200°F. and stabilizing for two minutes minimum. The thermometer shall then be removed from the liquid and vibrated in accordance with the following:

a. Mount the thermometer so that the dial face is either normal or parallel to the direction of vibration.

b. Subject the thermometer to two one-hour cycles of vibration (two hours total vibration time) using a minimum 1 "G" level. For each cycle the vibration frequency shall be progressively increased from 500 to 3000 cycles per minute during the first half hour and then progressively decreased from 3000 cycles per minute during the next half hour.

After vibration, another temperature reading shall be obtained by immersing the thermometer stem in liquid, having the same temperature as that used in the original reading, for two minutes minimum.

4.4.4 Dial crystal. The dial crystal shall be subjected to a temperature of 225°F. for a one-hour period.

* 4.5 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One thermometer, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Thermometers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Thermometers shall be dried in accordance with MIL-P-116.

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5.1.2.3 Preservative application. A preservative shall not be used.

5.1.2.4 Unit packaging. Each thermometer shall be packaged in accordance with MIL-P-116, method III. In addition, each thermometer shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.5 Intermediate package. Twelve thermometers, unit packaged as specified in 5.1.2.4, shall be packaged in an intermediate container conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Thermometers shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity..

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (144 thermometers) shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Thermometers shall be packed as specified in 5.2.1 except that the exterior container shall conform to PPP-B-636, class domestic. In addition, waterproofing and strapping shall not be required.

5.2.3 Level C. The packaged thermometers shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. In addition to any special markings specified in the contract or purchase order, each unit package, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

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6.2 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6685-00-444-6500	THERMOMETER, SELF-INDICATING, BIMETALLIC, Dial, 5-1/2 Inch

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet military requirements.

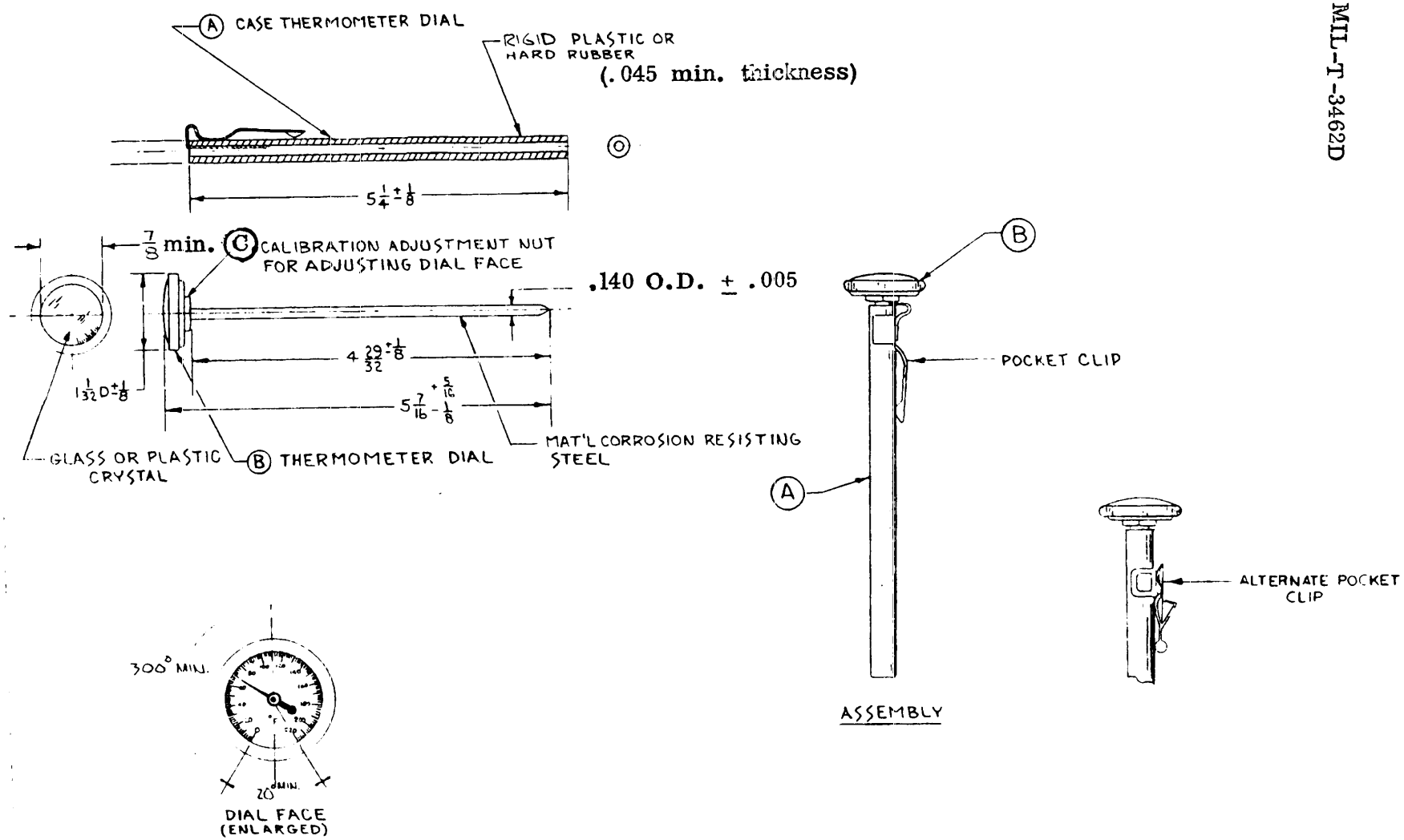
6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6685-0566

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THERMOMETER, SELF-INDICATING
BIMETALLIC, DIAL 5 1/2 INCH

FIGURE 1

Unless otherwise indicated, all dimensions are in inches.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-T-3462D

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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